



UNIVERSITÀ
DI TRENTO
Dipartimento di
Ingegneria Industriale

Master Course: Mechatronics Engineering
INTELLIGENT VEHICLES and AUTONOMOUS DRIVING

Autonomous Driving System for Traffic Light

Professor

Rosati Papini Gastone Pietro

Student

Murgia Edoardo, Mat: 247210

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1. Introduction

Objective:

Implementation of **longitudinal control** of an autonomous agent that manages traffic lights.

Human like behaviors:

- action priming → **minimal** intervention principle
(minimum square jerk criterion)

$$\begin{cases} \dot{s}(t) = v(t) \\ \dot{v}(t) = a(t) \\ \dot{a}(t) = j(t) \end{cases} \quad J = \int_0^T j(t)^2 dt$$

$$s(t) = c_1 t + \frac{1}{2} c_2 t^2 + \frac{1}{6} c_3 t^3 + \frac{1}{24} c_4 t^4 + \frac{1}{120} c_5 t^5 \quad ;$$

- action selection → basal ganglia structure.

